VNR Vignana Jyothi Institute of Engineering

Computer Science and Engineering | Course Code: CS303

Computer Networks & Internet Protocols - Laboratory Manual & Practical G

Syllabus Focus: CS303 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Physical & Data Link Protocol Foundations

" Syllabus Scope: Transmission media, framing, error detection CRC-32, Hamming code, sliding window flow

Unit 2: Medium Access & Local Area Networks

" Syllabus Scope: ALOHA, CSMA/CD Ethernet, CSMA/CA WiFi 802.11, VLANs, STP Spanning Tree Protocol, ARP pr

CS303: Computer Networks & Internet Protocols - Advanced Topics & Assessment

Academic Track: Computer Science and Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Network Layer & Routing Protocols

" IPv4 and IPv6 addressing, CIDR subnetting, Distance Vector Bellman-Ford, Link State Dijkstra/OSPF, Border Ga

Unit 4: Transport Layer & Congestion Control

" UDP datagrams, TCP 3-way handshake, connection teardown, TCP flow control, AIMD congestion control Slow Star

Unit 5: Application Layer Protocols & Network Security

" DNS resolution hierarchy, HTTP/1.1 vs HTTP/2 vs HTTP/3 QUIC, SMTP, TLS/SSL handshake, firewalls and NAT

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Packet Capture and Protocol Analysis of ARP, DNS, TCP Handshake using Wireshark
Lab 2: Socket Programming in C/Python: Multi-Threaded Chat Server using TCP and UDP Sockets
Lab 3: Network Topology Simulation and Routing Configuration using Cisco Packet Tracer
Lab 4: Sliding Window Protocol Simulation Go-Back-N and Selective Repeat under Packet Loss

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* James F. Kurose, Keith W. Ross - Computer Networking: A Top-Down Approach Pearson 8th Ed
* Andrew S. Tanenbaum, David J. Wetherall - Computer Networks Pearson 5th Ed

Authorized NoteVault Educational Document " Verified for Registered University Students